

Classification of age-related changes in lumbar intervertebral discs: 2002 Volvo Award in basic science.

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STUDY DESIGN: A histologic study on age-related changes of the human lumbar intervertebral disc was conducted. OBJECTIVES: To investigate comprehensively age-related temporospatial histologic changes in human lumbar intervertebral disc, and to develop a practicable and reliable classification system for age-related histologic disc alteration. SUMMARY OF THE BACKGROUND DATA: No comprehensive microscopic analysis of age-related disc changes is available. There is no conceptual morphologic framework for classifying age-related disc changes as a reference basis for more sophisticated molecular biologic analyses of the causative factors of disc aging or premature aging (degeneration). METHODS: A total of 180 complete sagittal lumbar motion segment slices obtained from 44 deceased individuals (fetal to 88 years of age) were analyzed with regard to 11 histologic variables for the intervertebral disc and endplate, respectively. In addition, 30 surgical specimens (3 regions each) were investigated with regard to five histologic variables. Based on the semiquantitative analyses of 20,250 histologic variable assessments, a classification system was developed and tested in terms of validity, practicability, and reliability. The classification system was applied to cadaveric and surgical disc specimens not included in the development of the classification system, and the scores were assessed by two additional independent raters. RESULTS: A semiquantitative analyses provided clear histologic evidence for the detrimental effect of a diminished blood supply on the endplate, resulting in the tissue breakdown beginning in the nucleus pulposus and starting in the second life decade. Significant temporospatial variations in the presence and abundance of histologic disc alterations were observed across levels, regions, macroscopic degeneration grades, and age groups. A practicable classification system for age-related histologic disc alterations was developed, resulting in moderate to excellent reliability (kappa values, 0.49-0.98) depending on the histologic variable. Application of the classification system to cadaveric and surgical specimens demonstrated a significant correlation with age (< 0.0001) and macroscopic grade of degeneration (< 0.0001). However, substantial data scatter caution against reliance on traditional macroscopic disc grading and favor a histology-based classification system as a reference standard. CONCLUSIONS: Histologic disc alterations can reliably be graded based on the proposed classification system providing a morphologic framework for more sophisticated molecular biologic analyses of factors leading to age-related disc changes. Diminished blood supply to the intervertebral disc in the first half of the second life decade appears to initiate tissue breakdown.